YASSIN SOLIMAN

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Education

Schulich School of Engineering (University of Calgary) Calgary, AB
Bachelor of Science in Software Engineering *Expected April 2028*

- **Relevant Coursework/Hobbies**: Operating Systems, Computer Architecture & Organization, Databases, Digital Circuits, Embedded Systems, Full-stack Web Development, Networks, Object-Oriented Programming, Data Structures & Algorithms, Software Design, Statistics, Machine Learning & Deep Learning, Software Architecture, Testing, Gaming, Homelabbing, GPU, Interpersonal skills, Artificial Intelligence (AI), Automation

Technical Skills

Languages: C, C++, Python, Java, SQL (MySQL), JavaScript, HTML5, CSS, TypeScript, TailwindCSS, Arduino
Frameworks and Libraries: React, NextJS, Three.js, JUnit, Swing, Django, NodeJS, NumPy, OpenCV, Ultralytics YOLOv8, supervision (ByteTrack), AWS (API Gateway, Lambda, DynamoDB, SNS), Platform Tools, CI/CD
Developer Tools and Environments: Atlassian, Jira, Codex, Cursor, Git, GitHub, Visual Studio Code, PyCharm, IntelliJ, Xcode, Cygwin, AutoCAD (CAD), MPLAB X IDE, Docker, Kubernetes, macOS, Windows 11, Linux, Unix, Shell Scripting, VMWare Fusion, VirtualBox, UTM, SSH, Proxmox VE, TrueNAS, Tailscale, GitHub Workflows

Experience

Software Lead Nov. 2024 – Present
Waybionic Club | | | ***Arduino, C, C++, Next.js, TypeScript, TailwindCSS, Agile*** *Calgary, AB*

- Led development of control systems for a **robotic surgical arm**, programming in **C, C++, and Arduino** to achieve precise and reliable motion control
- Integrated **sensors, actuators, and real-time embedded algorithms**, ensuring responsive performance of robotic components; additionally developed **Next.js/TypeScript** website to showcase and document the project
- Directed a cross-functional team of 8 engineers through **Agile** sprints and code reviews, improving productivity by **40%**

Software Engineer May 2024 – May 2025
Little Footprints | | ***JavaScript, HTML, CSS, REST APIs, Agile*** *Calgary, AB*

- Developed fintech websites using **JavaScript, HTML, and CSS** with **REST API** integration, boosting sales by **30%**
- Administered data-driven social media strategies, **increasing engagement by 70%**
- Enhanced frontend performance and **SEO** through web traffic analysis and SSL integration, enabling **50% more users** to visit
- Collaborated in **Agile** workflows with 2 software engineers, refining iterative solutions and communication

Programming Instructor July 2023 – Sept. 2023
Code Ninjas | | ***Python, Data Structures, Cryptography, Curriculum Design, Robotics*** *Calgary, AB*

- Taught core programming and **data structures** concepts using interactive games and robotics projects, expanding student engagement by **75%**
- **Guided 160+ students** through software development challenges and iteratively refined curriculum based on feedback, boosting coding confidence and troubleshooting skills by **30%**

Projects

Yassin.app | | | **Next.js, React, TypeScript, TailwindCSS, Three.js, Docker, Proxmox VE** Dec. 2025

- Designed an interactive, OS-style personal portfolio with draggable windows and app-like navigation using **Next.js, React, TypeScript, TailwindCSS, and Three.js** to showcase projects and experience
- Containerized the application with **Docker** and deployed it on a self-hosted **Proxmox VE** homelab, achieving reliable uptime and responsive performance without relying on third-party PaaS
- Implemented **GitHub**-driven **CI/CD** workflows to automatically build, test, and deploy changes on push, enabling rapid experimentation with minimal downtime

PathGuard | | | **Python, YOLOv8, ESP32-CAM, Arduino, AWS, React, TypeScript** Nov. 2025

- Built a crosswalk safety prototype combining an **ESP32-CAM** people-counting node, an **Arduino UNO R4 WiFi** hazard kiosk, and an **AWS** backend to surface real-time pedestrian congestion and hazards on a map
- Implemented a **YOLOv8n** + ByteTrack pipeline to detect and track pedestrians from an MJPEG stream, computing 5-minute rolling averages and publishing compact "low/med/high" congestion records to an ingest API
- Programmed a joystick-driven **LCD** UI and ultrasonic pedestrian detector with 9 hazard categories, sending structured `self_report` events to **DynamoDB** and triggering **SNS** email alerts